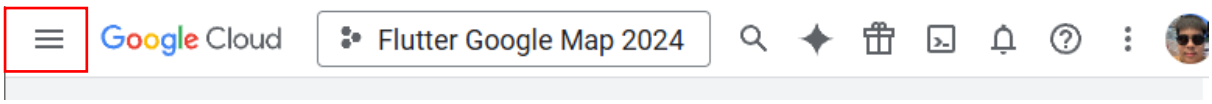
Google Map

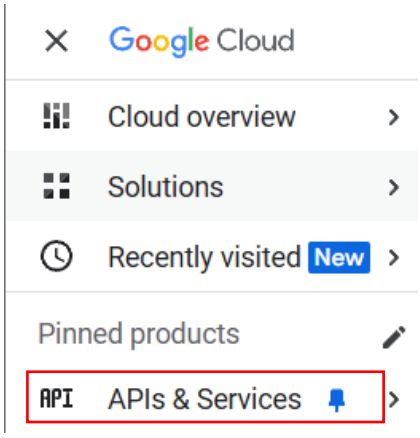
https://console.cloud.google.com/

- Login ด้วยอีเมลของมหาลัย

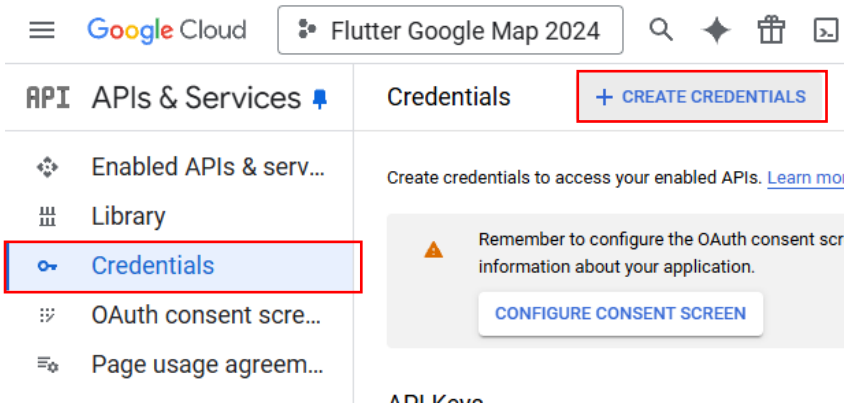
1. คลิก 3 ขีด



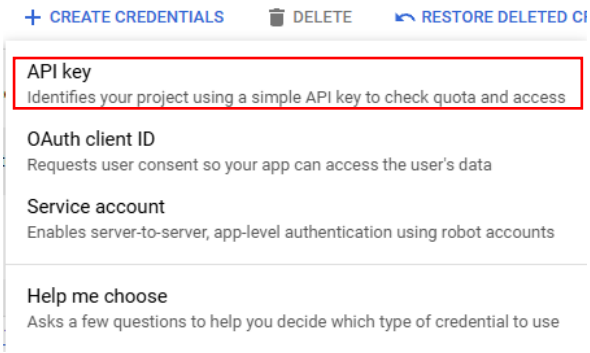
2.คลิก APIs & Services



3. คลิก Credentails และ + CREATE CREDENTIALS



4.คลิก API key ระบบจะสร้างคีย์มาให้

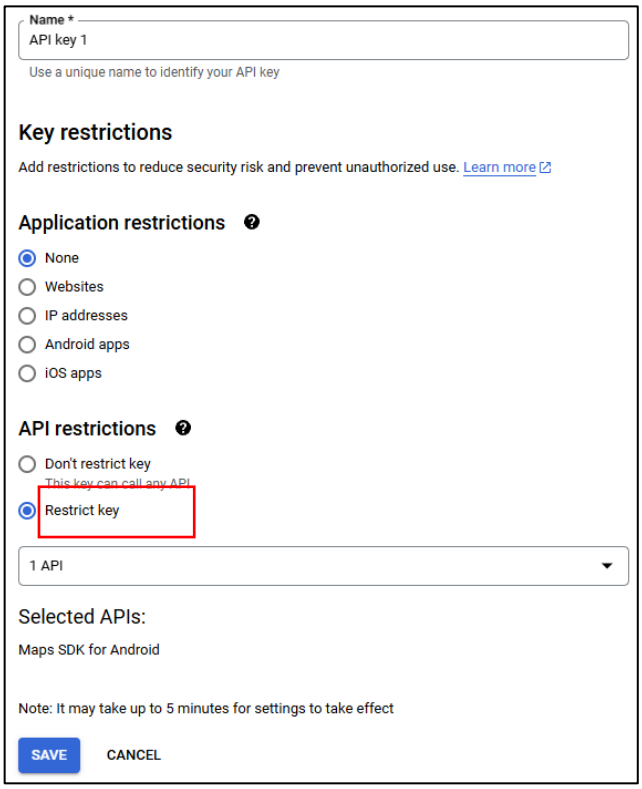


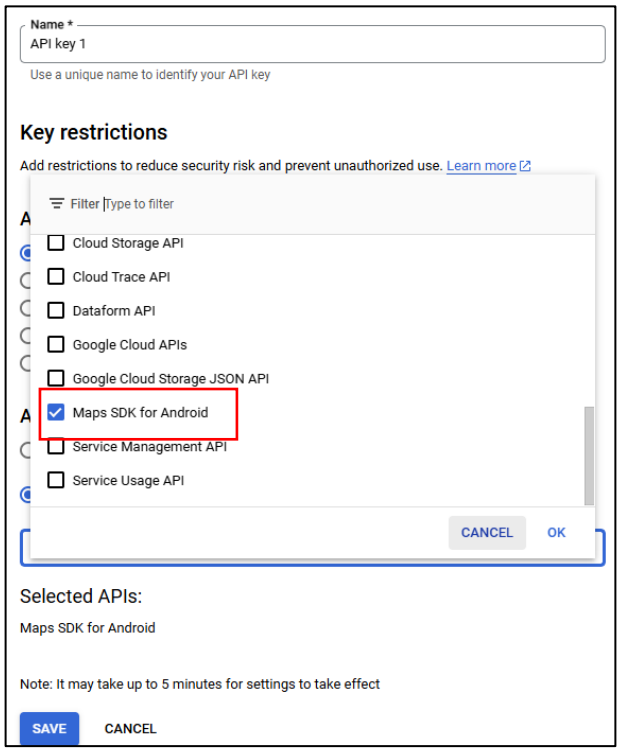
5. คลิกเข้าไปที่ API key 1

API Keys

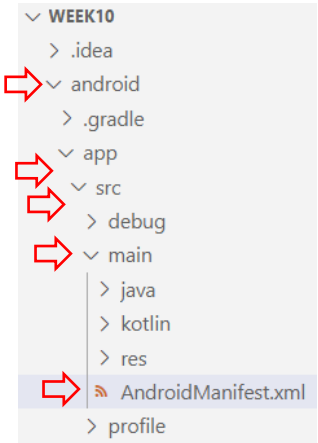
Name	Creation date	Restrictions	Actions
API key 1	Feb 27, 2025	Maps SDK for Android	SHOW KEY

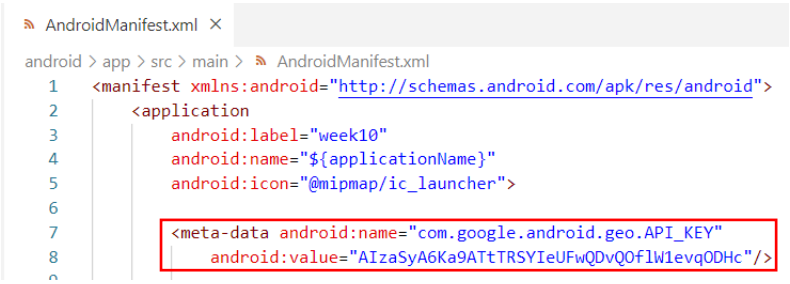
- กำหนดค่าให้กับ API Key 1 และ SAVE





6. ไปที่ \android\app\src\main\AndroidManifest.xml และเพิ่มคำสั่งในวงสี่เหลี่ยม





7.ติดตั้ง google_maps_flutter และ กำหนดเวอร์ชั่น ของ SDK ไปที่ android/app/build.gradle

AndroidManifest.xml | pubspec.yaml

pubspec.yaml

```yaml
name: week10
description: "A new Flutter project."
publish_to: 'none'
version: 1.0.0+1
environment:
  sdk: ^3.6.1
dependencies:
  flutter:
    sdk: flutter
  cupertino_icons: ^1.0.8
  google_maps_flutter: ^2.10.0
  geolocator: ^13.0.2

dev_dependencies:
  flutter_test:
    sdk: flutter
  flutter_lints: ^5.0.0
flutter:
  uses-material-design: true
```

WEEK10
- .idea
- android
  - .gradle
  - app
    - src
    - build.gradle

AndroidManifest.xml | build.gradle

android > app > build.gradle

```gradle
android {

    defaultConfig {
        applicationId = "com.example.week10"
        minSdk = 21
        targetSdk = flutter.targetSdkVersion
        versionCode = flutter.versionCode
        versionName = flutter.versionName
    }
```

8.

- lib
  - screens
    - current_location_screen.dart
    - simple_map_screen.dart
  - home_screen.dart
  - main.dart

main.dart

```dart
import "package:flutter/material.dart";
import "package:google_maps_flutter/google_maps_flutter.dart";
import "home_screen.dart";
void main() => runApp(MyApp());
class MyApp extends StatelessWidget {
  const MyApp({super.key});
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: const HomeScreen(),
    );
  }
}
```

home_screen.dart

```
1  import 'package:flutter/cupertino.dart';
2  import 'package:flutter/material.dart';
3  import 'screens/simple_map_screen.dart';
4  import 'screens/current_location_screen.dart';
5  class HomeScreen extends StatefulWidget {
6    const HomeScreen({super.key});
7    @override
8    State<HomeScreen> createState() => _HomeScreenState();
9  }
10 class _HomeScreenState extends State<HomeScreen> {
11   @override
12   Widget build(BuildContext context) {
13     return Scaffold(
14       appBar: AppBar(
15         title: const Text("Google Map New "),
16         centerTitle: true,
17       ),
18       body: SizedBox(
19         width: MediaQuery.of(context).size.width,
20         child: Column(
21           children: [
22             ElevatedButton(
23                 onPressed: () {
24                   Navigator.of(context)
25                       .push(MaterialPageRoute(builder: (BuildContext context) {
26                     return MapSample();
27                   }));
28                 },
29                 child: const Text("Simple Map")),
30             ElevatedButton(
31                 onPressed: () {
32                   Navigator.of(context)
33                       .push(MaterialPageRoute(builder: (BuildContext context) {
34                     return CurrentLocation();
35                   }));
36                 },
37                 child: const Text("Current Location")),
38           ],
39         ),
40       ),
41     );
42   }
43 }
```

Screens / simple_map_screen.dart

```
import 'dart:async';
import "package:flutter/material.dart";
import "package:google_maps_flutter/google_maps_flutter.dart";
class MapSample extends StatefulWidget {
  const MapSample({super.key});
  @override
  State<MapSample> createState() => _MapSampleState();
}
class _MapSampleState extends State<MapSample> {
  final Completer<GoogleMapController> _controller = Completer();
  static const CameraPosition _kGooglePlex = CameraPosition(
    target: LatLng(37.42796133580664, -122.085749655962),
    zoom: 14.0,
  );
  static const CameraPosition _kLake = CameraPosition(
    bearing: 192.8334901395799,
    target: LatLng(37.42796133580664, -122.085749655962),
    tilt: 59.44,
    zoom: 19.0,
  );
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      body: GoogleMap(
        mapType: MapType.hybrid,
        initialCameraPosition: _kGooglePlex,
        onMapCreated: (GoogleMapController controller) {
          _controller.complete(controller);
        },
      ),
      floatingActionButton: FloatingActionButton.extended(
        onPressed: _goToTheLake,
        label: const Text("Go to the Lake"),
        icon: const Icon(Icons.directions_boat),
      ),
    );
  }
  Future<void> _goToTheLake() async {
    final GoogleMapController controller = await _controller.future;
    controller.animateCamera(CameraUpdate.newCameraPosition(_kLake));
  }
}
```

Screens / current_location_screen.dart

```dart
import 'package:flutter/material.dart';
import 'package:geolocator/geolocator.dart';
import "package:google_maps_flutter/google_maps_flutter.dart";
class CurrentLocation extends StatefulWidget {
  const CurrentLocation({super.key});

  @override
  State<CurrentLocation> createState() => _CurrentLocationState();
}
class _CurrentLocationState extends State<CurrentLocation> {
  late GoogleMapController googleMapController;
  static const CameraPosition _initialCameraPosition =
      CameraPosition(target: LatLng(14.1593588, 101.3463206), zoom: 16);
  Set<Marker> markers = {};
  final LatLng _posi1 = const LatLng(14.1651265, 101.34394);
  final LatLng _posi2 = const LatLng(14.1589163, 101.3453382);
  final LatLng _posi3 = const LatLng(14.1650378, 101.3377752);
  void _onMapCreated(GoogleMapController controller) {
    googleMapController = controller;
    _addMarker(_posi1, 'Hopak boy');
    _addMarker(_posi2, 'Digital');
    _addMarker(_posi3, 'Airport');
  }
  void _addMarker(LatLng position, String markerId) {
    setState(() {
      markers.add(Marker(
          markerId: MarkerId(markerId),
          position: position,
          infoWindow: InfoWindow(title: markerId)));
    });
  }
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text("User Current Location"),
        centerTitle: true,
      ),
      body: GoogleMap(
        markers: markers,
        mapType: MapType.hybrid,
        initialCameraPosition: _initialCameraPosition,
        onMapCreated: _onMapCreated,
      ),
    );
  }
  Future<Position> _determinePosition() async {
    bool serviceEnabled;
    LocationPermission permission;
    serviceEnabled = await Geolocator.isLocationServiceEnabled();
    if (!serviceEnabled) {
      return Future.error("Location services are disabled.");
    }
    permission = await Geolocator.checkPermission();
    if (permission == LocationPermission.denied) {
      permission = await Geolocator.requestPermission();
      if (permission == LocationPermission.denied) {
        return Future.error("Location permission denied.");
      }
    }
    if (permission == LocationPermission.deniedForever) {
      return Future.error("Location permission permanently denied.");
    }
    return await Geolocator.getCurrentPosition();
  }
}
```